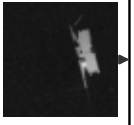
Training (synthetic images only)



GT crop, (q, t)

3 views / image
clean
photometric + in-plane roll θ
$q' = \mathrm{Exp}(-\theta d) \otimes q$
$t' = R(d, -\theta)\, t$
d: crop-centre ray

DINOv3-L (frozen)
8×8×1024 tokens

MoGe-2 (frozen)
normals 16×16×4

Tesseract tree Transformer
4 charts + 5 levels × 8 children
soft targets, 2.4 M params

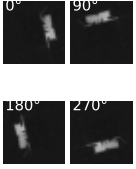
Tesseract tree Transformer
same architecture, own weights

Translation block
SPACE-HOP target u
(optional, 6-DoF)

Loss
Σ_ℓ CE(level ℓ)
$+ \|u - u^*\|^2$

Orbit distillation (optional)
teacher: orbit over the 3 views
student loss $L_{GT} + \beta H(\text{teacher} \to v)$

Inference: orbit voting over 4 in-plane views



Per view k
frozen backbones + trees
per-level PoE:
$\log p_k \propto \log p_D + \log p_M$ (per level)
beam search → 8 leaves
(DINOv3-only: no MoGe-2)

Map back
$c = \mathrm{Exp}(+\theta_k d) \otimes c_k$
4 views × 8 = 32
candidate rotations
(exact inverse of the
training label transform)

Orbit score
$s(c) = \Sigma_k \log p_k(\mathrm{Exp}(-\theta_k d) \otimes c)$
tree likelihood of each candidate
as seen from every view
$\hat{q} = \arg\max_c s(c)$
(alt.: weighted vote, 5° kernel)

Output
$\hat{q}$: leaf centre
$\hat{t}$: block u per view,
back-rolled and averaged

frozen: 634 M params
trained: 2.4 / 4.5 M

a look-alike (e.g. 180° flip) rarely wins in all rotated views; the true pose, mapped back, is consistent across them